

# **Perception-Aware Chance-Constrained Model Predictive Control (PAC-MPC)**

**Integrating control decisions and action-dependent  
perception uncertainty for constrained autonomous  
operation**

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# Introduction & Motivation

- Autonomous systems operate in uncertain, dynamic environments requiring active perception via sensors.
- Traditional MPC assumes perception is independent of control, neglecting physical sensing limits like field-of-view and range.
- Perception quality directly depends on control actions such as motion trajectory, sensor orientation, and beamforming.
- PAC-MPC closes the loop between control actions and perception uncertainty to achieve less conservative trajectories.

flowchart LR

```
A[Control Action  $x_s, u_s$ ] -->|Governs FOV & Range| B[Perception Quality]
B -->|Reduces Estimation Covariance| C[Constraint Tightening  $\gamma$ ]
C -->|Expands Feasible Region| D[Less Conservative Trajectory]
```

# System, Environment & Estimator Modeling

- Known deterministic system dynamics:  $xs_{k+1} = fs(xs_k, us_k)$  subject to admissible sets  $X$  and  $U$ .
- Dynamic environment state:  $xe_{k+1} = fe(xe_k, \psi_k)$  modeling stationary obstacles or moving agents.
- Action-dependent measurements:  $ye_k = qe(xe_k, \zeta_k, xs_k, us_k)$  dependent on system state and inputs.
- Fixed estimator provides mean  $\mu_e$  and covariance  $\Sigma_e$  updates driven by system states and control actions.

| Model Component | Variable / State               | Governing Equation          | Key Characteristics |
|-----------------|--------------------------------|-----------------------------|---------------------|
| System Model    | $xs \in R^{nx}, us \in R^{nu}$ | $xs_{k+1} = fs(xs_k, us_k)$ | Known deterministic |

# Perception-Aware Chance Constraints Formulation

- Safety constraints enforced as Individual Chance Constraints (ICCs):  $P[hs_l | x_s + he_l | x_e \leq hb_l] \geq 1 - \epsilon_l$ .
- Deterministic reformulation converts chance constraints into tightened bounds via backoff parameter  $\gamma_{\text{bar}}(\Sigma^e)$ .
- Environment Predictor computes predicted mean  $\mu^e$  and covariance  $\Sigma^e$  using measurement prediction error  $\Sigma_y$ .
- PAC-MPC optimizes augmented state  $\xi = (x_s, \Sigma_e)$  to stabilize tracking error while managing perception uncertainty.

```
graph TD
    A[Current State  $x_s, \mu_e, \Sigma_e$ ] --> B[Environment Predictor]
    B --> C[Compute Tightening  $\gamma_{\text{bar}}$ ]
    C --> D[Online Finite Horizon Optimal Control Backlog]
```

# Recursive Feasibility & Closed-Loop Stability

- Predictor Monotonicity (Assumptions 3 & 4): Ensures prediction bounds do not underestimate constraint tightening.
- Terminal Set Invariance: Parametrized terminal set  $Z_f(M_e)$  guarantees recursive feasibility in probability.
- Joint Cost Function: Merges control performance stage/terminal costs with perception stage/terminal costs.
- Lyapunov Stability: Guarantees asymptotic stability in probability for joint system state and estimation covariance  $\xi$ .

# **Constructive Design Procedure for Linear Dynamics**

# Constructive Design Procedure for Linear Dynamics

- Applies to linear system dynamics ( $A_s$ ,  $B_s$ ), linear estimator  $\Lambda$ , and polyhedral constraint sets.
- Terminal Controller: Linear feedback  $u = K_f x_s + G_f r$  constructs Maximum Constrained Admissible Set (MCAS)  $O_\infty$ .
- Perception Stage & Terminal Costs: Formulated via squared Frobenius norms of covariance error over horizon  $N_p$ .
- LMI Optimization: Terminal cost weight  $P_c$  and compensation matrix  $M_c$  designed iteratively via LMIs.

```
import numpy as np

def compute_pac_mpc_cost(xs, us, Sigma_e, rx, ru, Sigma_r, Qc, Rc, Sc, Pc, Wc, N_p, Lambda):
    # Control stage cost
```

# Case Study & Numerical Results

- Evaluated on benchmark double integrator system with input-dependent and state-dependent sensor quality.
- Closed-loop trajectories successfully regulate system state to setpoint while satisfying chance constraints.
- Monte Carlo Validation: 100 runs over 200 time steps yielded 98% empirical constraint satisfaction (exceeding 95% target).
- Real-Time Performance: Average solver execution time of 22 ms per step (worst-case < 30 ms) using IPOPT in CasADi.

| Parameter / Performance Metric | Value / Configuration                         |
|--------------------------------|-----------------------------------------------|
| System Benchmark Model         | Double Integrator (Sampling time $T_s = 1$ s) |
| Chance Constraint Threshold    | $1 - \varepsilon = 0.95$ (95% safety bound)   |

# Conclusions & Future Outlook

- PAC-MPC establishes a systematic methodology unifying perception uncertainty with predictive control decision-making.
- Guarantees recursive feasibility and closed-loop probabilistic stability for autonomous systems in unknown environments.
- Significantly reduces trajectory conservatism compared to passive estimation baselines.
- Future Work: Integration with learned neural perception models, specialized SQP solvers, and experimental testbeds.

# Key Takeaways

- Active Coupling: Controlling system motion directly influences perception quality and constraint tightness.
- Rigorous Safety: Guarantees recursive feasibility and closed-loop stability under probabilistic individual chance constraints.
- Real-Time Viability: Demonstrates 98% empirical safety and sub-30ms computation times for real-time robotic deployment.

**Keywords:** Model Predictive Control, Perception-Aware Control, Chance Constraints, Autonomous Vehicles, Uncertain Environments

# Thank You

Questions & Discussion